Fundamental theorems and Continuity Equation:-

(a) fundamental theorem of gradient :

$$\int_a^b (\vec{\nabla}\phi)\cdot\vec{dl} = \phi(b) - \phi(a)$$

(b) fundamental theorem of divergence

$$\int_{Vol} (\vec{\nabla}\cdot\vec{V})\, dv = \int_{surface} \vec{V}\cdot\vec{ds}$$

(c) fundamental theorem of curl :-

$$\int_{surface} (\vec{\nabla}\times\vec{V})\cdot\vec{ds} = \int_{line} \vec{V}\cdot\vec{dl}$$ (also called stoke's thm)

(d) Continuity Equation

$$\vec{\nabla}\cdot\vec{J} + \frac{\partial \rho}{\partial t} = 0$$

Electric field :- A region of space around a charge in which any other charge experiences force of attraction or repulsion is called electric field.

The electric field of a charge is measured in terms of vector quantity called Electric field Intensity $(\vec{E})$.

The electric field intensity of a charge at any given point (P) is defined as force acting on unit positive charge at that point.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

SI unit: N/C

for point charge q, electric intensity at distance 'r' is given by

$$\vec{E} = \left(\frac{1}{4\pi\epsilon_0\epsilon_r}\right)\frac{q}{r^2}\,\hat{r}$$

$\hat{r} \Rightarrow$ unit vector

In magnitude

$$E = \left(\frac{1}{4\pi\epsilon_0\epsilon_r}\right)\left(\frac{q}{r^2}\right)$$

Electric field due to a continuous charge distribution.

(a) **for line charge:**

$\because$ linear charge density $\lambda = \frac{dq}{dl}$

$\therefore dq = (\lambda)(dl)$

$\therefore q = \int_{line} \lambda \, dl$

$$\vec{E} = \frac{1}{4\pi\epsilon_0\epsilon_r} \int_{line} \frac{\hat{r}}{r^2} \lambda \, dl$$

(b) **for Surface charge:**

$\because$ surface charge density, $\sigma = \frac{dq}{ds}$

$\therefore dq = \sigma \, ds$

$q = \int_{surface} \sigma \, ds$

$$\therefore \vec{E} = \left(\frac{1}{4\pi\epsilon_0\epsilon_r}\right) \int_{Surface} \frac{\hat{r}}{r^2} \sigma \, ds$$

(c) **for Volume charge:-**

$\because$ volume charge density $\rho = \frac{dq}{dv}$

$\therefore dq = \rho \, dv$

$q = \int_{Vol} \rho \, dv$

$$\vec{E} = \left(\frac{1}{4\pi\epsilon_0\epsilon_r}\right) \int_{Vol} \frac{\hat{r}}{r^2} \rho \, dv$$